DO NOW — RECALL

# Lesson 084 · A* Search: g, h and f Scores

Name: _____ Class: _____ Date: _____

**1. What is Dijkstra's tentative distance?**

**2. What does f(n) equal in A*?**

**3. What is an admissible heuristic?**

**4. Binary search target12 in sorted [2,4,6,8,10,12,14]. Show midpoint indexes.**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

Known best route cost from source.

### 2. Model answer

g(n)+h(n).

### 3. Model answer

Never overestimates true remaining cheapest cost.

### 4. Model answer

mid3=8 then mid5=12; return index5.

### By the end, you can…

• A* orders candidate states using f(n)=g(n)+h(n), where g is known cost from the start and h estimates remaining cost to the goal.

• A heuristic that never overestimates remaining true cost is admissible; with suitable search implementation/assumptions this supports optimality.

• Track open candidates, their g/h/f and predecessors. An overestimating heuristic or incorrect update can make search miss the shortest path.

### Vocabulary — know these terms

A* — Heuristic-guided shortest-path search.

g(n) — Cost so far from start.

h(n) — Estimated cost to goal.

f(n) — g(n)+h(n) priority.

NEW LEARNING

# Read, annotate and explain

## Observe

A* orders candidate states using f(n)=g(n)+h(n), where g is known cost from the start and h estimates remaining cost to the goal.

## Apply

A heuristic that never overestimates remaining true cost is admissible; with suitable search implementation/assumptions this supports optimality.

## Evaluate

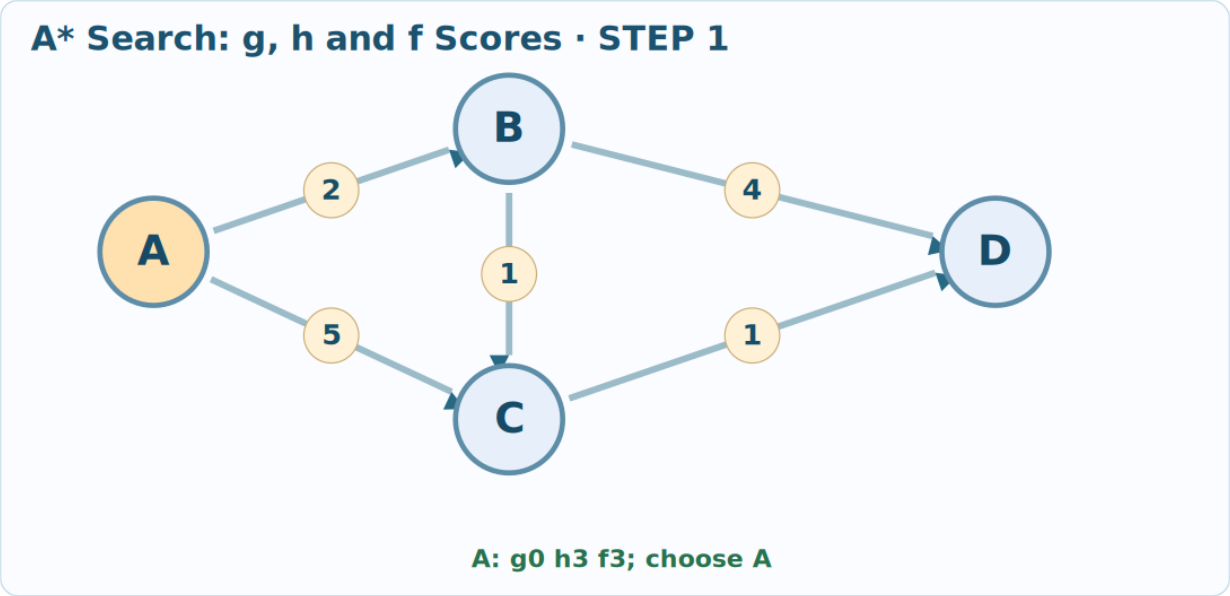
Track open candidates, their g/h/f and predecessors. An overestimating heuristic or incorrect update can make search miss the shortest path.

**Explain one step in your own words, using the correct technical terms.**

# New Learning · original lesson diagrams

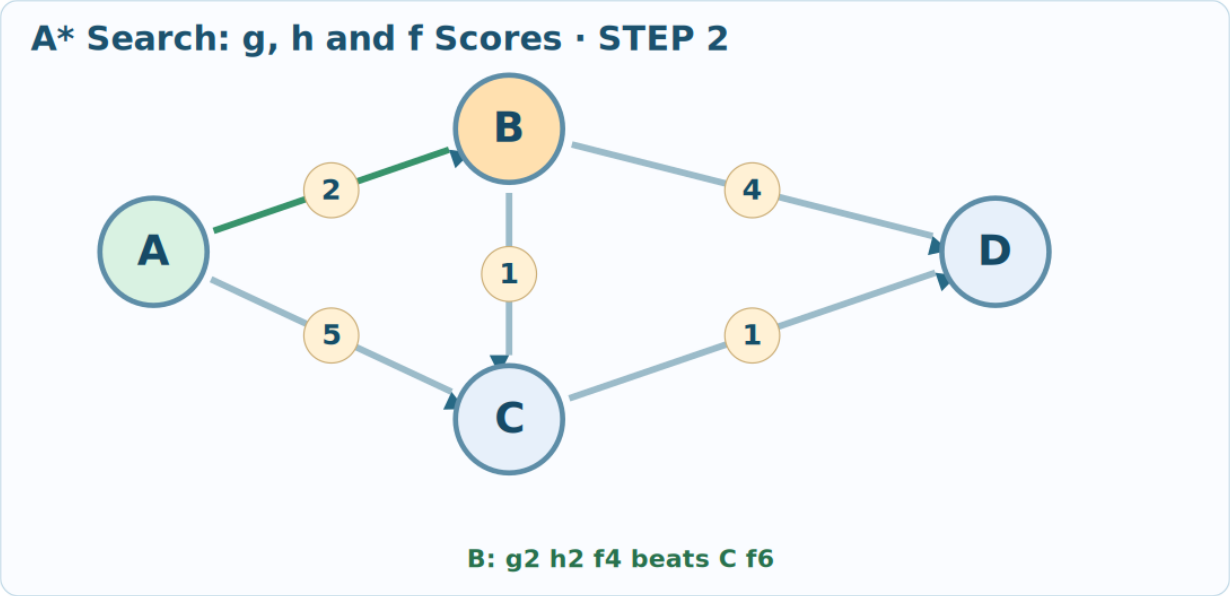
Diagrams appear after the related explanation. Annotate the original source visual.

### New Learning




A* Search: g, h and f Scores · STEP 1: A: g0 h3 f3; choose A

### New Learning

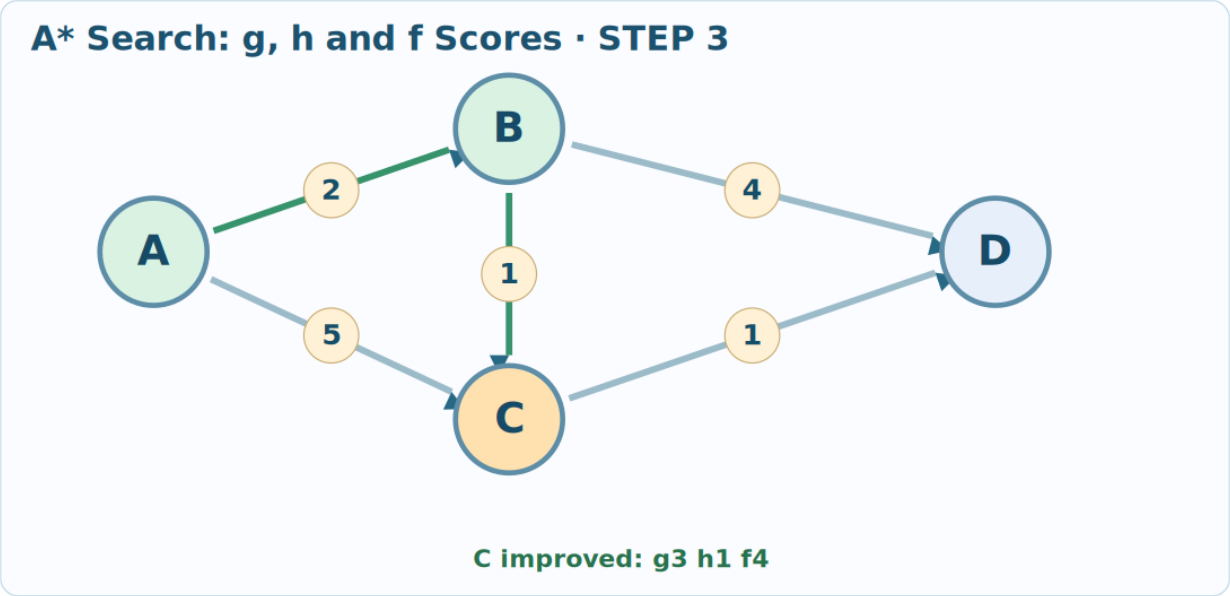



A* Search: g, h and f Scores · STEP 2: B: g2 h2 f4 beats C f6

# New Learning · original lesson diagrams

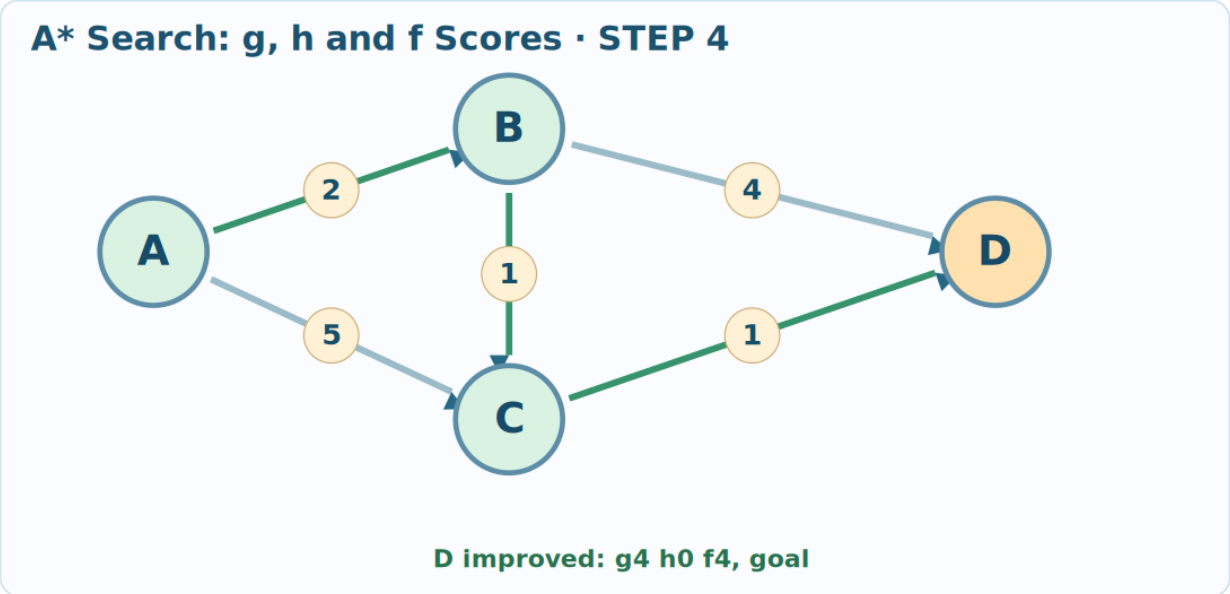
Diagrams appear after the related explanation. Annotate the original source visual.

### New Learning




A* Search: g, h and f Scores · STEP 3: C improved: g3 h1 f4

### New Learning




A* Search: g, h and f Scores · STEP 4: D improved: g4 h0 f4, goal

I DO — WORKED EXAMPLE

# Follow the worked method

### Find P→R: P→Q = 3, P→R = 8, Q→R = 2. Heuristics: h(P)=5, h(Q)=2, h(R)=0.

### Calculate all frontier f-scores

From P: Q has g=3,h=2,f=5 ; R has g=8,h=0,f=8 . Put both in OPEN. Choose Q next (smallest f), not simply the nearest-looking node.

### Improve the path to R

Via Q→R, new g(R)=3+2=5 , which improves the existing 8. Thus f(R)=5+0=5 , and predecessor R becomes Q. OPEN now contains R with the improved values.

### Confirm the path and heuristic meaning

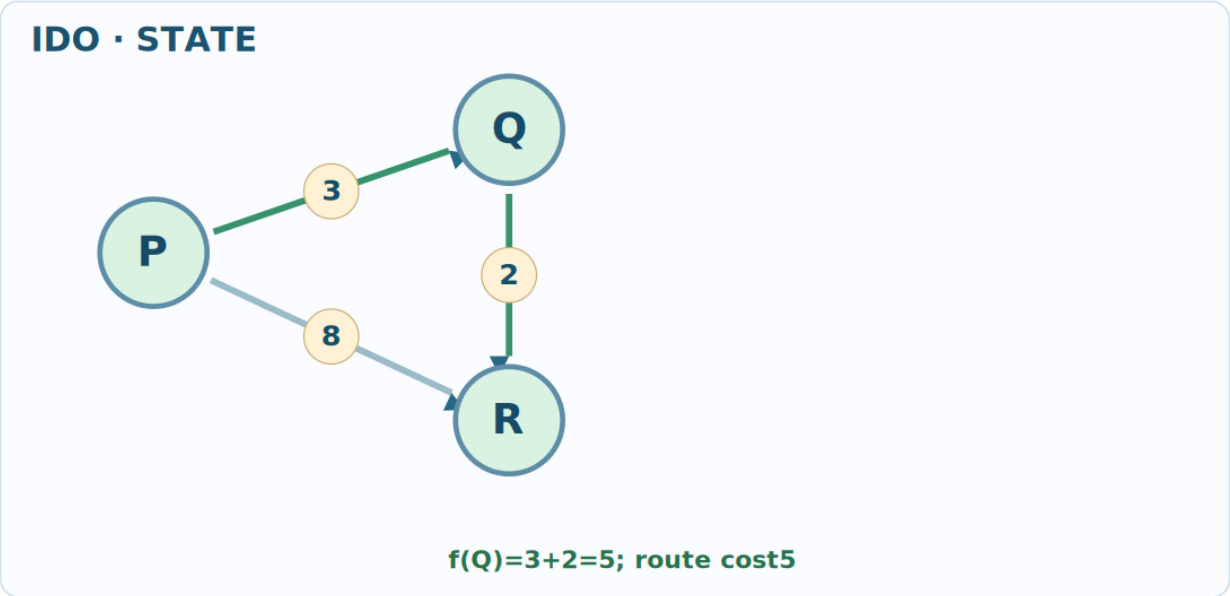
Select goal R from OPEN, then reconstruct R←Q←P to obtain P→Q→R , cost 5 . The supplied heuristics do not overestimate remaining true cost; the next lesson can explore what happens when this fails.

# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Find P→R: P→Q = 3, P→R = 8, Q→R = 2. Heuristics: h(P)=5, h(Q)=2, h(R)=0.**

From P: Q has g=3,h=2,f=5 ; R has g=8,h=0,f=8 . Put both in OPEN. Choose Q next (smallest f), not simply the nearest-looking node.




IDO · STATE: f(Q)=3+2=5; route cost5

WE DO — GUIDED PRACTICE

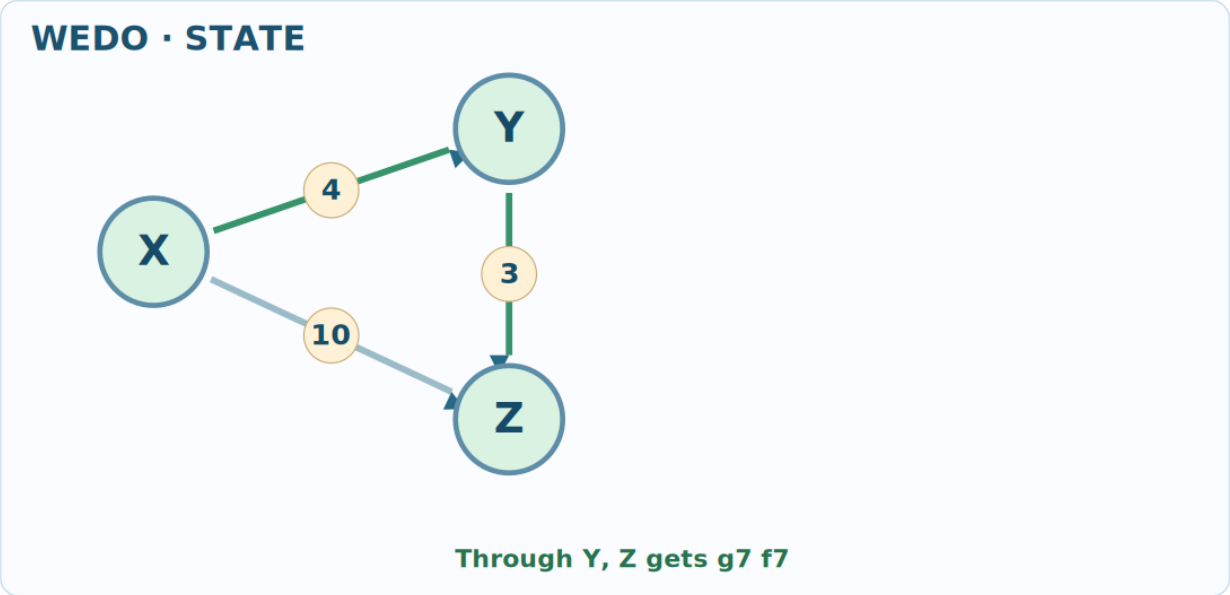
# Try → Discuss → Improve

# We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Find X→Z: X→Y = 4, X→Z = 10, Y→Z = 3. h(X)=7, h(Y)=3, h(Z)=0.**

Calculate the two g/f pairs from the edges leaving X. State which candidate A* should expand next and why.




WEDO · STATE: Through Y, Z gets g7 f7

# WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

**Guided 1: Compute priorities and choose Calculate the two g/f pairs from the edges leaving X. State which candidate A* should expand next and why.**

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

After expanding A, calculate f for B and C using stated graph/heuristics. [6]

### Question 2

After expanding B then C, show improvements to g(C) and g(D); state best path. [6]

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Compare A* with Dijkstra when h=0 everywhere, and explain risk of overestimation. [6]

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**A* h(n) is the exact distance to the goal for every possible heuristic.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

## GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

# WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

### Guided 1 — original CHECK

Expand Y

Y has g=4,h=3,f=7 , Z has g=10,h=0,f=10 , so expand Y. Via Y→Z, candidate g is 4+3=7: improve Z from 10 to 7, with predecessor Z←Y.

Finish the trace

Now Z has g=7,h=0,f=7 . Select it as the goal; predecessor chain Z←Y←X gives X→Y→Z , cost 7 . Distinguish the actual path cost g from estimated priority f.

IMPROVE: Explain one correction to your own reasoning.